

Cardiac Risk After Heart-Sparing Breast Radiotherapy

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IMPORTANCE Radiotherapy for breast cancer exposes the heart to incidental radiation, and historical data have shown a dose-dependent increase in associated cardiac events. Although whole-heart dose metrics are commonly used for risk assessment, emerging evidence suggests that dose to the left anterior descending coronary artery (LAD) may better capture risk.

OBJECTIVE To compare the ability of heart and LAD radiation dose metrics to predict cardiac risk after breast radiotherapy.

DESIGN, SETTING, AND PARTICIPANTS This was a cross-sectional study of patients with breast cancer who were treated with either 3-dimensional conformal or intensity-modulated radiotherapy from 2008 to 2018 at a tertiary care center in Canada. The primary analysis included patients with left-sided breast cancer. Cardiac events were assessed using longitudinal follow-up data. Dosimetry was derived from computed tomography plans using automated segmentation and converted to equivalent dose in 2-Gy fractions (EQD₂). Data were analyzed from September 2024 to April 2026.

MAIN OUTCOMES AND MEASURES Adverse cardiac events defined as myocardial infarction or hospital admission or emergency department visit for unstable angina (ie, acute coronary syndrome), arrhythmia, heart failure, pericarditis, or myocarditis. The incidence of coronary angiography and coronary revascularization was also captured as coronary artery disease (CAD). Discrimination for dose metrics was performed with receiver operator characteristic curves and competing-risks regression (Fine and Gray), adjusted for cardiovascular risk factors.

RESULTS The analysis included 4908 patients with breast cancer of whom 2223 had left-sided breast cancer. During a median (IQR) follow-up period of 10.8 (8.4-13.1) years, cumulative incidence of cardiac event or CAD was 5.0 (95% CI, 4.1-6.0) at 10 years. A data-driven cut point analysis identified 12 Gy EQD₂ as the maximum LAD dose that best stratified risk. Among patients with left-sided breast cancer, maximum LAD dose (concordance [C] index, 0.58; 95% CI, 0.52-0.64) discriminated better than mean heart dose (C index, 0.53; 95% CI, 0.47-0.60). In multivariable analysis, maximum LAD of 12 Gy or greater was independently associated with higher cardiac risk (subdistribution hazard ratio = 1.81; 95% CI, 1.04-3.16; *P* = .04), whereas mean heart dose of 2 Gy or greater was not associated (*P* = .99). For clinical context, the 12-Gy EQD₂ for maximum LAD dose corresponds to a physical dose of approximately 10.5 Gy for 42.5 Gy in 16 fractions and 7 Gy for 26 Gy in 5 fractions.

CONCLUSIONS AND RELEVANCE In this cross-sectional study of heart-sparing breast radiotherapy, LAD dose was associated with cardiac events, whereas whole-heart metrics was not. These findings support LAD-based planning and respiratory motion management to reduce long-term cardiovascular risk in patients with breast cancer undergoing radiotherapy.

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For patients with breast cancer, radiotherapy is a standard component of treatment that reduces the risk of local recurrence.^{1,2} However, delivering breast radiotherapy can be associated with incidental radiation of nearby organs, including the heart.³ Prior studies⁴⁻⁶ have established a dose-response relationship indicating that radiation dose to the heart during breast radiotherapy is associated with an increased risk of major adverse cardiac events (MACEs). Although these studies largely evaluated cardiac doses that were considerably higher than doses reported by modern radiotherapy techniques, they continue to influence survivorship care expectations for patients undergoing modern breast radiotherapy.^{7,8}

As a standard in breast radiotherapy, the whole heart structure is used in dose evaluation and reporting.⁹ Recent evidence suggests that radiation dose to specific cardiac substructures, especially the left anterior descending coronary artery (LAD), correlates more strongly with predicting cardiac outcomes or that the whole-heart dose is an inadequate predictor altogether.^{6,8,10-12} This is not unique to breast radiotherapy; in thoracic tumors, similar investigations have focused on substructure-specific risks.¹³⁻¹⁶

In response to evidence associating cardiac dose to morbidity, substantial updates to treatment techniques, including respiratory motion management, have been directed toward limiting incidental heart exposure.¹⁷⁻²⁴ These modern doses are substantially lower than the doses that were previously validated in establishing the relationship to MACE as well as typical heart mean dose reports of 2 to 4 Gray (Gy).^{9,25,26} Although the implicit clinical value of reducing cardiac dose is widely acknowledged, there remains limited direct evidence confirming an association between lower doses and measurable reductions in long-term cardiac morbidity²⁷; therefore, strategies to reduce cardiac dose have not been universally adopted.²⁷

Among cardiac substructures, the LAD has a central role in radiation-associated MACE. However, prospective LAD contouring remains uncommon.^{23,28,29} Although recommended dose limits to the LAD have been adopted in the National Comprehensive Cancer Network (NCCN) Guidelines for Hodgkin Lymphoma,³⁰ similar clinically actionable objectives have not been established for breast cancer radiotherapy in North America. The dose limits adopted by the NCCN were based on studies with cardiac doses that averaged an order of magnitude higher than the doses typically reported for modern breast radiotherapy, which challenges the suitability of adopting these guidelines for breast cancer.¹⁴

In this study, we analyzed a population-based cohort of patients with breast cancer treated with heart-sparing radiotherapy to evaluate the association between radiation dose to the LAD and adverse cardiac events. We compared this association with the conventional whole-heart dose metrics to assess the predictive discrimination of the LAD and whole heart in capturing risk.

Methods

This study was reviewed and approved by the University of Calgary Ethics Review Board (HREBA.CC-22-00242), which also waived informed consent.

Key Points

Question In patients with breast cancer treated with contemporary heart-sparing breast radiotherapy, is radiation dose to the left anterior descending coronary artery (LAD) associated with more long-term adverse cardiac events compared with conventional whole-heart dose metrics?

Findings This cross-sectional study of 4908 patients with breast cancer (2223 left-sided) found that higher maximum LAD dose had a statistically significant independent association with increased risk of cardiac events, whereas the whole-heart dose did not.

Meaning These findings suggest that LAD-specific dosimetry may better identify cardiac risk than whole-heart metrics in modern breast radiotherapy, and supports incorporating LAD objectives into plan quality evaluation.

Study Population

For all patients in this study, provincial registry systems captured cardiac outcomes, comorbidities, and all-cause mortality. For radiotherapy, all patients treated at a tertiary care center in southern Alberta (Calgary, Canada) were included in the study with data extracted from the radiation oncology information system.³¹ This cancer center is the designated radiotherapy facility for the regional population in Canada, which minimizes referral selection bias. The catchment area for the radiotherapy center is approximately 1.5 million of the 4 million provincial inhabitants. For all patients, the date of first radiotherapy fraction was extracted and used to define the date of pre- and postevent.

Patient Data Collection, Cardiac Outcomes, and Comorbidities

The incidence of adverse cardiac event, preexisting comorbidities using previously validated codes from the *International Statistical Classification of Diseases, Tenth Revision, Clinical Modification (ICD-10-CM)* and all-cause mortality were extracted from Alberta Vital Statistics via patients' electronic medical records.^{32,33} An adverse cardiac event was defined as 1 or more of either myocardial infarction or admissions or emergency department presentation for unstable angina (grouped as acute coronary syndrome), heart failure, arrhythmia, pericarditis, or myocarditis.³⁴ Comorbidities included hypertension, hyperlipidemia, diabetes, and chronic kidney disease. The report by Southern et al³³ provides the complete list of *ICD-10* codes for cardiac events and comorbidities.

The incidence of coronary angiography and coronary revascularization were determined through the prospective Alberta Provincial Project for Outcome Assessment in Coronary Heart disease (APPROACH).³⁵ Initiated in January 1995 using the *Ninth Revision* of the *ICD-CM*, the APPROACH registry was designed to study outcomes of care for patients with coronary artery disease (CAD).^{36,37} Obstructive CAD was defined as stenosis greater than or equal to 50% in the left main or greater than or equal to 70% in other coronary arteries.^{38,39} This study evaluated the distribution of obstructive CAD in the LAD, left circumflex, or right coronary artery (RCA) territories. A preexisting history was defined as any event documented before the first fraction of radiotherapy.

Radiotherapy Treatment Technique

From 2008 to 2018, patients with breast cancer were eligible for inclusion if treated with 3-dimensional (3-D) conformal radiotherapy or intensity-modulated radiotherapy (IMRT).⁴⁰ The initial time point for inclusion (2008) was the introduction of 3-D computed tomography (CT)-based radiotherapy planning, allowing for Digital Imaging and Communications in Medicine files to be used on patient-specific anatomy. Beginning in 2013, patients were eligible for treatment using the deep inspiration breath hold technique (DIBH). To account for differences in fractionation between patients, all radiation doses were converted to equivalent dose in 2-Gy fractions (EQD₂) where the α/β ratio was taken to be 3 for late cardiac effects.⁴¹

Heart and LAD Segmentation

Heart and LAD segmentations were generated using a validated artificial intelligence (AI)-based deep learning model.^{42,43} The LAD contour was defined using an 8-mm diameter brush proximally at the bifurcation of the left main into the LAD and left circumflex and extending along the course of the LAD through the anterior interventricular groove with the distal end at the ventricular apex.⁴⁴ AI-based segmentation has been reported to improve the fidelity of LAD contours, addressing prior evidence that prospectively generated manual clinical contours were less consistent.^{42,43}

Statistical Analysis

Descriptive statistics were used to summarize baseline clinical and treatment characteristics. Categorical variables were reported as frequencies and percentages, while continuous variables were presented as median (IQR).

To eliminate treatment laterality as a confounding factor, the primary analysis of outcomes was performed on the left-sided cohort in EQD₂, with additional cohort analyses provided in eTables 1 and 3 in Supplement 1. Follow-up from the start of radiotherapy was estimated using the Kaplan-Meier method.

For time-to-event analyses, time was initiated at the first delivered fraction of radiotherapy and concluded at the earliest of (1) first qualifying cardiac event, (2) death, or (3) last available follow-up. The latest possible adjudication date was July 2024. Cumulative incidence is reported as the composite of CAD or cardiac event as well as these events separately, adjusting for death as a competing risk.

To assess the discriminative performance of radiation dose metrics, receiver operating characteristics (ROC) curve analysis was conducted for both mean heart dose (MHD), and mean and maximum LAD dose in EQD₂. For the maximum LAD, the area under the curve (AUC) was computed to evaluate the relative predictive accuracy. Cut points for binary classification were identified using the Liu method,⁴⁵ which assesses discriminative performance by the maximum of the product of sensitivity and specificity. This exploratory cut point selection investigated a clinically relevant range spanning 7 to 20 Gy EQD₂ and was used to generate a clinically interpretable value that could be evaluated for its association with cardiovascular events.^{8,9} For the mean whole-heart dose, a 2-Gy threshold was selected to be consistent with clinical guidelines.^{9,26} For the mean LAD dose, a 3-Gy threshold was

selected for comparison to the outcomes of a retrospective study.⁸

To identify factors associated with cardiac events after breast radiotherapy, univariable and multivariable competing risks regression models were constructed using the Fine and Gray method. Covariates evaluated included known cardiovascular prognostic factors, such as age, preexisting comorbidities, as well as radiotherapy treatment factors, including the use of DIBH, inclusion of supraclavicular field, boost delivery, and use of accelerated partial breast irradiation. Dose-response was assessed by modeling cardiac dose variables as continuous covariates (per Gy). For interpretability, the dichotomized parameterizations from the cut point analysis were evaluated in alternative model specifications to avoid collinearity. To assess the potential for nonlinearity in the dose-response between the maximum LAD dose (EQD₂) and cardiac events, restricted cubic splines with 4 knots at fifth, thirty-fifth, sixty-fifth, and ninety-fifth percentiles, and Fine and Gray regression models were evaluated.

Primary multivariable modeling included all variables with $P < .05$ in univariable analysis, along with key dosimetric variables, to assess their independent association with outcomes. For the left-sided DIBH subgroup model, only age, preexisting cardiovascular risk, and LAD maximum were included in a multivariable model to reduce overfitting. Multicollinearity was evaluated using variance inflation factor and tolerance thresholds. Hazard ratios (HRs), subdistribution hazard ratios (sHRs), and 95% CIs were reported.

Adjusted cumulative incidence curves were generated. Dose cut point stratified analyses were designed to assess the presence of clinically relevant objectives that could inform dosimetric aims in radiotherapy planning. Additionally, we compared the proportion of patients with left-sided breast cancer meeting dosimetric thresholds before and after routine implementation of DIBH using the χ^2 test of independence.

All analyses were performed using Stata/SE, version 17.0 (Stata Corp) from September 2024 to April 2026. Tests were 2-sided and statistical significance was defined as $P < .05$.

Results

The analysis included a total of 4908 patients with breast cancer who were treated with radiotherapy, among whom 2223 (45.3%) had left-sided breast cancer. DIBH was used in 559 patients (11.4%) of the total participants, and 465 with left-sided cancer (20.9%). Baseline clinical factors and treatment characteristics are reported in Table 1. A complete report of prescription utilization by year of radiotherapy is provided in the eFigure in Supplement 1.^{46,47} With a median (IQR) follow-up of 10.5 (8.2-13.0) years, CAD and adverse cardiac events occurred in 100 patients (2.0%) and 180 patients (3.7%), respectively, for a total of 225 patients (4.6%) having 1 or both.

Patient Population With Left-Sided Breast Cancer and Cardiac Event Rates

For the left-sided only cohort, which defined the cohort for primary analysis, and the median (IQR) follow-up was 10.8 years

Table 1. Clinical and Treatment Characteristics of Patients With Breast Cancer, Overall, by Disease Site, and by Subgroup Treated With Deep Inspiration Breath Hold (DIBH), 2008 to 2018

Characteristic	Total cohort, No. (%)		DIBH cohort, No. (%) ^a	
	Overall	Left-sided disease	Overall	Left-sided disease
Patients, No.	4908 (100)	2223 (45.3)	559 (11.4)	465 (20.9)
Age, median (IQR), y	59.6 (50.7-68.0)	59.9 (51.0-68.0)	57.4 (48.7-66.5)	58.3 (49.8-67.3)
Follow-up, median (IQR), y	10.5 (8.2-13.0)	10.8 (8.4-13.1)	7.1 (6.2-8.2)	7.2 (6.3-8.3)
Cardiovascular medical history				
Hypertension	1109 (22.7)	509 (22.9)	105 (18.9)	91 (19.6)
Hyperlipidemia	311 (6.4)	143 (6.4)	23 (4.1)	19 (4.1)
Diabetes	411 (8.4)	188 (8.5)	40 (7.2)	33 (7.1)
Chronic kidney disease	85 (1.7)	46 (2.1)	9 (1.6)	7 (1.5)
Coronary artery disease	25 (0.5)	13 (0.6)	5 (0.9)	4 (0.9)
Acute coronary syndrome	12 (0.3)	5 (0.2)	1 (0.2)	1 (0.2)
Heart failure	4 (0.1)	4 (0.2)	0	0
Arrhythmia	86 (1.8)	41 (1.9)	8 (1.4)	8 (1.7)
Breast cancer laterality				
Right-sided breast cancer	2609 (54.0)	NA	93 (16.7)	NA
Left-sided breast cancer	2197 (45.5)	2197 (98.8)	464 (83.2)	464 (83.2)
Bilateral	26 (0.5)	26 (1.2)	1 (0.2)	1 (0.2)
Radiotherapy target				
Breast only	2253 (45.9)	1054 (47.0)	158 (28.3)	156 (33.6)
Breast + breast boost	774 (15.8)	339 (15.3)	109 (19.5)	106 (22.8)
Breast + SCV	530 (10.8)	227 (10.2)	75 (13.4)	56 (12.0)
Breast + breast boost + SCV	264 (5.4)	108 (4.9)	56 (10.0)	37 (8.0)
Chest wall only	36 (0.7)	20 (0.9)	6 (1.1)	6 (1.3)
Chest wall + breast boost	2 (0)	0	0	0
Chest wall + SCV	790 (16.1)	359 (16.2)	148 (26.5)	100 (21.5)
Chest wall + breast boost + SCV	19 (0.4)	6 (0.3)	5 (0.9)	3 (0.7)
APBI	213 (4.3)	93 (4.2)	1 (0.2)	1 (0.2)
Radiotherapy prescription dose				
40-42.5 Gy in 16 fractions	3683 (75.1)	1638 (73.7)	512 (91.6)	430 (92.5)
45-50.4 Gy in 25-28 fractions	591 (12.0)	288 (13.0)	41 (7.3)	30 (6.4)
27 Gy in 5 fractions	183 (3.7)	82 (3.7)	1 (0.2)	0
Other	451 (9.2)	215 (9.6)	5 (0.9)	5 (1.1)
DIBH utilization (2013-2018)				
2013	NA	NA	35	35
2014	NA	NA	49	47
2015	NA	NA	36	34
2016	NA	NA	102	87
2017	NA	NA	166	138
2018	NA	NA	171	124
EQD ₂ adjusted dose				
Mean heart dose, median (IQR) Gy	0.4 (0.2-0.7)	NA	0.6 (0.4-0.7)	NA
Right-sided breast cancer	0.2 (0.2-0.3) ^a	NA	0.3 (0.3-0.4) ^a	NA
Left-sided breast cancer	0.7 (0.6-1.2) ^a	0.7 (0.6-1.2)	0.6 (0.5-0.7) ^a	0.6 (0.5-0.7)
Mean LAD dose, median (IQR) Gy	0.1 (0.1-2.5)	NA	1.8 (1.3-2.3)	NA
Right-sided breast cancer	0.1 (0.0-0.1) ^a	NA	0.1 (0.1-0.1) ^a	NA
Left-sided breast cancer	2.7 (1.8-8.4) ^a	2.7 (1.8-8.4)	1.9 (1.6-2.3) ^a	1.9 (1.6-2.3)
Maximum LAD dose, median (IQR) Gy	0.3 (0.2-15.2)	NA	5.8 (3.5-10.1)	NA
Right-sided breast cancer	0.2 (0.1-0.2) ^a	NA	0.2 (0.2-0.3) ^a	NA
Left-sided breast cancer	18.9 (5.7-43.7) ^a	18.9 (5.7-43.7)	6.7 (4.7-11.6) ^a	6.7 (4.7-11.6)

Abbreviations: APBI, accelerated partial breast irradiation; CAD, coronary artery disease; EQD₂, equivalent dose in 2-Gy fractions; Gy, Gray; LAD, left anterior descending coronary artery; NA, not applicable; SCV, supraclavicular.

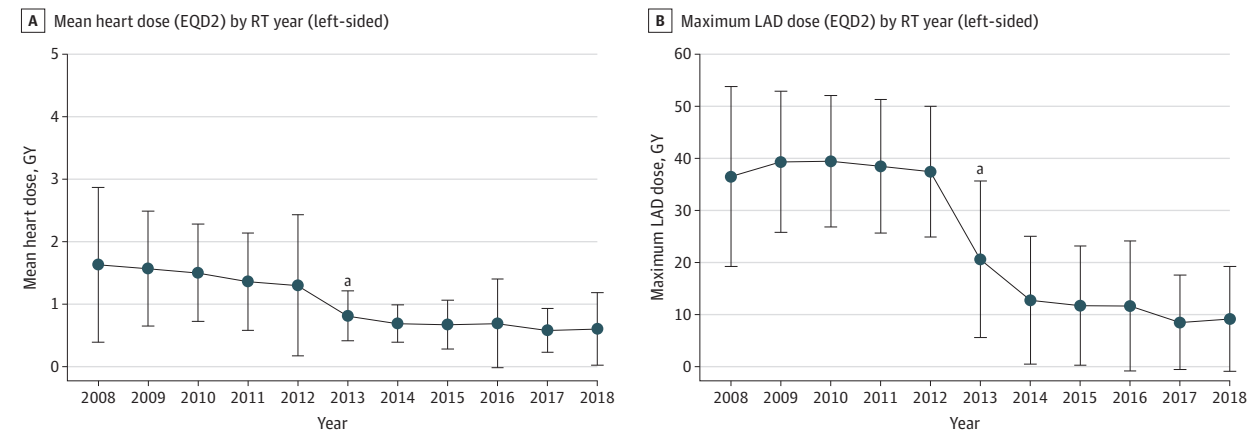
^a $P < .001$ in comparing patients with right- and left-sided breast cancer for mean heart dose, mean LAD dose, and maximum LAD dose.

Table 2. Cardiac Event Types and Cumulative Incidence Rates for Left-Sided Breast Cancer Cohort Only

Cardiac event type	Patients, No. (%)	Time to event, median (IQR), y ^a	Cumulative incidence (95% CI)		
			5 y	10 y	15 y
Any CAD or cardiac event	115 (5.2)	4.5 (1.9-7.8)	2.8 (2.2-3.6)	5.0 (4.1-6.0)	6.2 (5.1-7.4)
CAD	57 (2.6)	4.5 (1.9-7.8)	1.4 (0.9-1.9)	2.6 (1.9-3.3)	3.0 (2.3-3.8)
Cardiac event	91 (4.1)	3.9 (1.8-7.3)	2.4 (1.8-3.1)	4.0 (3.2-4.9)	4.9 (3.9-6.0)
Acute coronary syndrome	36 (1.6)	2.3 (1.2-7.4)	1.0 (0.6-1.5)	1.5 (1.0-2.1)	2.0 (1.4-2.8)
Heart failure	35 (1.6)	5.4 (3.0-8.4)	0.7 (0.4-1.2)	1.6 (1.1-2.2)	1.9 (1.3-2.6)
Arrhythmia	30 (1.4)	4.6 (2.4-7.1)	0.7 (0.4-1.2)	1.3 (0.9-1.9)	1.6 (1.1-2.4)
Pericarditis	2 (0)	NA	NA	NA	NA
Myocarditis	1 (0)	NA	NA	NA	NA
Mean heart dose ≥ 2 Gy	9 (4.6)	1.9 (1.5-7.8)	3.1 (1.3-6.2)	4.1 (1.9-7.6)	4.7 (2.3-8.4)
Mean heart dose < 2 Gy	82 (4.1)	4.1 (1.8-7.1)	2.3 (1.7-3.1)	4.0 (3.1-5.0)	4.9 (3.9-6.1)
Maximum LAD dose ≥ 12 Gy	67 (5.2)	4.2 (1.8-7.8)	2.8 (2.0-3.8)	4.6 (3.6-5.9)	5.6 (4.4-5.9)
Maximum LAD dose < 12 Gy	24 (2.6)	3.7 (1.4-6.1)	1.8 (1.1-2.8)	3.0 (1.9-4.4)	3.0 (1.9-4.4)

Abbreviations: CAD, coronary artery disease; Gy, Gray; LAD, left anterior descending coronary artery; NA, not applicable.

^a Time to event was calculated descriptively among patients who developed a cardiac event.

Figure 1. Dot and Whisker Plot of Mean Equivalent Dose in 2-Gy Fractions (EQD₂) for Mean Heart and Maximum Left Anterior Descending Coronary Artery (LAD) per Year of Radiotherapy (RT) for Patients With Left-Sided Breast Cancer

^aImplementation of the deep inspiration breath hold technique (DIBH) in 2013 is noted. Whiskers indicate 1 SD.

(8.4-13.1), the median (IQR) age at treatment was 59.9 (51.0-68.0) years. Either CAD ($n = 57$) or an adverse cardiac event ($n = 91$) occurred in a total of 115 patients (5.2%), accounting for patients with both event types, with a median (IQR) time to event of 4.5 (1.9-7.8) years. The cumulative incidence was 5.0 (95% CI, 4.1-6.0) during the 10-year follow-up period. A complete report on cardiac event types and cumulative incidence is provided in Table 2, with overall cohort outcomes and right-sided cohort outcomes reported in eTable 1 in Supplement 1, respectively.

Changes in Dose to the Heart and LAD Over Time

For the left-sided cohort, the mean doses in EQD₂ for the mean heart and maximum LAD by year of radiotherapy are shown in Figure 1. Linear regression supports a statistically significant change in maximum LAD dose across years. From 2013 through 2017, DIBH utilization in patients with left-sided breast cancer was 15.5%, 22.5%, 13.3%, 45.8%, and 89.7%, respectively, and ultimately in 2018, 90.5%.

Dose-Response Association and Cardiac Events

For patients with left-sided breast cancer, on AUROC analysis, maximum LAD dose (EQD₂) of 12 Gy had the highest discriminative performance (concordance [C] index, 0.58) and was selected for inclusion in subsequent regression models (eTable 2 in Supplement 1 shows the complete evaluation of C index for other maximum LAD doses). Maximum LAD dose outperformed both mean heart dose (C index, 0.53) and mean LAD dose (C index, 0.54).

In univariable Fine and Gray analysis for patients with left-sided breast cancer, the 12-Gy maximum LAD dose threshold was associated with an increased risk of cardiac event (sHR, 1.81; 95% CI, 1.04-3.16; $P = .04$), whereas the 2-Gy mean heart dose and 3-Gy mean LAD dose were not (Table 3 [$P > .05$]). After adjusting for known cardiovascular risk factors and radiotherapy factors, the maximum LAD dose of 12 Gy or greater was associated with an increased risk of cardiac events, indicating an independent dosimetric effect. The characterization of patients by any of the radiotherapy treatment factors was

Table 3. Fine and Gray Analysis of Competing Risks Regression Associated With Cardiac Events for Left-Sided Breast Cancer Cohort Only

Variable	Univariable analysis		Multivariable analysis	
	HR (95% CI)	P value	sHR (95% CI)	P value
Age	1.08 (1.06-1.10)	< .001	1.07 (1.04-1.09)	< .001 ^a
Preexisting medical factors				
Hypertension	2.95 (1.96-4.46)	<.001	1.35 (0.84-2.14)	.21
Hyperlipidemia	2.22 (1.21-4.10)	.01	1.26 (0.63-2.50)	.51
Diabetes	2.24 (1.29-3.90)	.004	1.36 (0.79-2.34)	.27
Chronic kidney disease	6.69 (3.48-12.86)	<.001	3.10 (1.52-6.32)	.002 ^a
Coronary artery disease	9.75 (3.53-26.89)	<.001	2.54 (1.10-5.86)	.03 ^a
Cardiac event prior to RT	11.71 (6.81-20.12)	<.001	4.89 (2.58-9.26)	<.001 ^a
Radiotherapy treatment factors				
DIBH	0.48 (0.23-1.00)	.05	0.74 (0.33-1.68)	.48
Any boost	0.50 (0.26-0.97)	.04	0.88 (0.44-1.75)	.71
Any SCV field	0.86 (0.54-1.36)	.52	NA	NA
Any APBI	0.97 (0.31-3.07)	.96	NA	NA
RT dose parameters (EQD ₂)				
MHD (Gy) ^b	1.21 (0.97-1.50)	.09	NA	NA
LAD Dmean (Gy) ^b	1.01 (1.00-1.02)	.13	NA	NA
LAD Dmax (Gy) ^b	1.00 (0.97-1.03)	.95	NA	NA
MHD ≥2 Gy (vs <2 Gy)	1.00 (0.50-2.00)	.99	0.90 (0.41-2.01)	.81
LAD Dmean ≥3 Gy (vs <3 Gy)	1.08 (0.71-1.63)	.73	NA	NA
LAD Dmax ≥12 Gy (vs <12 Gy)	1.70 (1.06-2.71)	.03	1.81 (1.04-3.16)	.04 ^a

Abbreviations: APBI, accelerated partial breast irradiation; CAD, coronary artery disease; DIBH, deep inspiration breath hold technique; Dmean, mean dose; Dmax, maximum dose; EQD₂, equivalent dose in 2-Gy fractions; Gy, Gray; HR, hazard ratio; LAD, left anterior descending coronary artery; MHD, mean heart dose; NA, not applicable; RT, radiotherapy; SCV, supraclavicular;

sHR, subdistribution hazard ratio.

^a Statistical significance on multivariable analysis.

^b Continuous dosimetric variables omitted in the multivariable analysis model due to collinearity with dichotomous variables.

not significant in multivariable models, reinforcing that cardiac risk is more directly captured by dose metrics than by treatment-related surrogates. In an additional model restricted to patients with left-sided cancer who used the DIBH technique (eTable 3 in Supplement 1), the 12-Gy dose cut point retained its statistical significance, indicating the outcome was not limited to treatment era.

The restricted cubic spline modeling did not demonstrate statistically significant evidence of nonlinearity in the dose-response for maximum LAD dose and cardiac events ($P = .28$). These findings support modeling the maximum LAD dose as a continuous predictor and the 12-Gy objective is interpreted as an empirically derived, clinically interpretable benchmark.

Time to Event Analysis (Cox Regression)

Figure 2 shows the cumulative incidence of cardiac events (excluding CAD) among patients due to 2 representative and impactful preradiotherapy comorbidities (panels A and B) as well as 2 cardiac dose parameters—mean heart dose and maximum LAD dose (panels C and D). Statistical significance is reported in Table 3.

Clinical Utility

The proportion of patients with left-sided breast cancer receiving a maximum LAD less than 12 Gy increased from 8.6% in the pre-DIBH era to 53.2% in the post-DIBH era ($\chi^2 P < .001$).

By contrast, in the pre-DIBH era, 55.7% of patients with left-sided breast cancer received a mean heart dose less than 2 Gy. Once DIBH became a standard option, 95.3% of patients received a mean heart dose less than 2 Gy ($\chi^2 P < .001$).

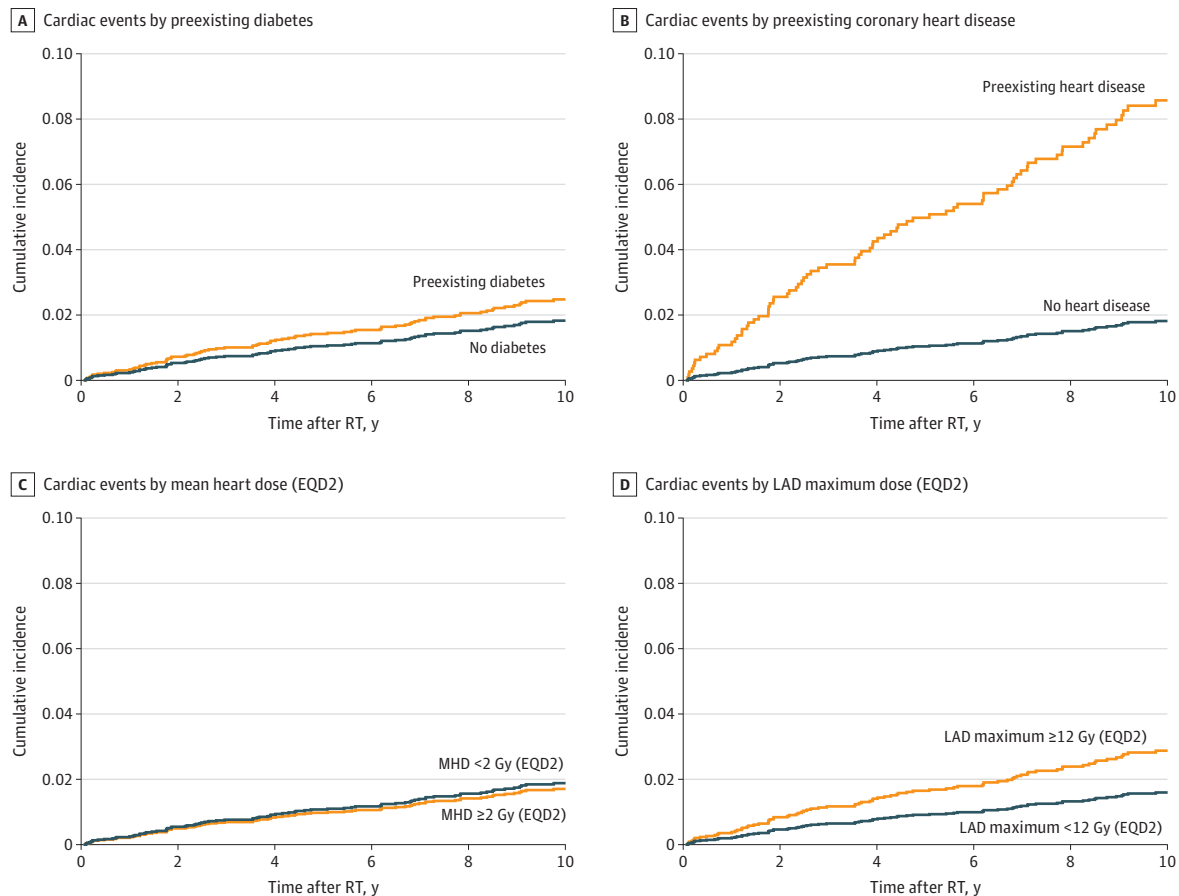
For commonly used prescription dose and fractionation regimens, the 12-Gy EQD₂ maximum LAD dose corresponds to a physical dose of approximately 10.5 Gy for 42.5 Gy in 16 fractions and 7 Gy for 26 Gy in 5 fractions. These are provided for clinical context and should not be interpreted as independently validated physical-dose thresholds.

Discussion

In this cross-sectional population-based cohort study of patients with breast cancer and individual coronary dosimetry, the maximum dose to the LAD was independently associated with the risk of adverse cardiac events, whereas whole-heart metrics were not. This indicates that the traditional reliance on whole-heart metrics may obscure clinically relevant injury patterns. A clinically relevant benchmark of 12 Gy (EQD₂) in maximum LAD dose was identified.

These results support a shift in focus from whole-heart dosimetry toward LAD-specific objectives during breast radiotherapy planning. Clinically, LAD sparing informs both patient-specific treatment decision-making and survivorship care, similar to strategies used in risk prediction for patients with

Figure 2. Cumulative Incidence Curves for Selected Factors Associated With Cardiac Events, Excluding Coronary Artery Disease, at Time From Radiotherapy (RT)



Cumulative incidence is stratified by the presence of 2 representative and impactful preradiation therapy comorbidities (A and B) and 2 cardiac dose parameters, mean heart dose and maximum LAD dose (C and D). EQD₂, mean

equivalent dose in 2-Gy fractions; LAD, left anterior descending coronary artery; MHD, mean heart dose.

lymphoma or esophageal cancer.^{48,49} This study's large, regionally representative cohort, comprehensive linkage to centralized outcome registries, and AI-assisted substructure dosimetry increases confidence in the observed dose-response relationship. Competing risks methods account for deaths that preclude subsequent cardiac events, yielding more accurate cumulative incidence estimates and strengthening inferences about the association between LAD dose and cardiac events when competing mortality is present. However, this method does not eliminate other sources of bias such as residual confounding. From a methodologic perspective, this work demonstrates the importance of collecting substructure dosimetry in prospective studies.

These findings extend the work of previous studies by showing that in a modern cohort, the association with cardiac events is more apparent for LAD-specific dosimetry than for whole-heart metrics.^{2,5,8} The sensibility of the cumulative incidence event rates is also supported by other studies, such as RADCOMP,⁵⁰ which was designed to evaluate whether proton therapy reduces major cardiovascular events in patients receiving regional nodal irradiation. The 10-year incidence (5.0%) in this study is similar

to the 3.5% to 6.3% estimated in RADCOMP.⁵⁰ Together, these findings underscore the importance of incorporating LAD dose into radiotherapy planning to alert treating physicians when patients are at elevated cardiac risk.

The implications of this work for clinicians reinforce the value of conscientious LAD sparing in breast radiotherapy, even for patients with satisfactory traditional whole-heart dosimetry. In addition, these findings suggest that LAD dose should be audited for breast radiotherapy treatments to evaluate the quality of the radiation oncology program. Overall, the central finding represents a clinically meaningful cardiovascular risk factor that should alert the treating physician to better manage long-term risk and that may be used to guide treatment optimization and patient surveillance.

Limitations

Limitations include the absence of tumor stage and systemic therapy details in our dataset, which prevented stratified analyses by these important clinical factors, and residual confounding remains possible. Systemic therapies, such as anthracyclines or *ERBB-2* targeted agents, can increase cardiovascular risk and

may modify the effect of radiotherapy; without these data we cannot fully separate their contribution from the radiation-related associations observed.⁵¹ Nonetheless, the pattern of association—particularly the specific association with maximum LAD dose rather than mean whole-heart dose—is consistent with mechanistic expectations for focal coronary injury and supports the clinical relevance of the finding.³⁴ Providing a dosimetry objective for the LAD is consistent with practice in radiotherapy where single-dose thresholds are applied across a cohort and individual risk factors are considered on a patient-by-patient basis. In addition, cardiac constraints may be achieved without reducing target coverage by drawing attention to radiation distribution patterns in treatment planning approaches.²³ However, the modest predictive performance of the maximum LAD should serve as a caution in clinical decision-making, especially in relationship to balancing target coverage goals.

The data-driven approach to multivariable model building carries a risk of overfitting and inflated false-positive findings. To mitigate these risks, we prespecified major clinical confounders such as age, prior CAD, and other comorbidities, and restricted automated selection to variables that were signifi-

cant on univariable analysis. Although these steps cannot eliminate the possibility of overfitting, they reduce its likelihood; replication or validation in an external cohort would further strengthen confidence in the association. This study also only evaluated the cardiac risk with the LAD and whole heart and did not exclude other substructures investigated in previous studies¹⁰ for association with risk.

Conclusions

This cross-sectional study found that the whole-heart dose is not associated with cardiac risk at low doses, whereas the LAD was independently associated with cardiac events. These findings substantiate the use of LAD dosimetry and strengthen the evidence associating cardiac irradiation with late cardiac morbidity, which supports a shift to including LAD-specific dose objectives in treatment planning. Incorporating LAD sparing into breast radiotherapy protocols and auditing LAD dose as part of institutional quality assurance may reduce preventable cardiac harm.

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